

DAVID WAGNER

Theoretical Physics Student | Statistical Mechanics & Open Quantum Systems

Theoretical physicist student specialized in analytical and numerical methods used in both equilibrium and non-equilibrium statistical physics. Seeking a final year internship starting between June & September 2026.

CONTACT

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📍 Living in Stockholm

SKILLS

Technical Skills

Programming Languages

Python	Advanced (Qiskit, Numpy, Scipy, Matplotlib)
C/C++	Intermediate
MATLAB	Advanced
SQL	Novice

Hardware and Experiments

Optical setups (Michelson, FTIR spectrometer, Goniometer, etc.)
Electrical circuits and control systems
Measurement and data analysis

Scientific Softwares

Git, version control & testing (pytest)
CAD (Autodesk Fusion 360, Creo Parametric)
L^AT_EX

Soft skills

Analytical and abstract thinking
Autonomy
Teamwork in research environments
Scientific rigor

Languages

French	mother tongue
English	C1 660/677 on TOEFL ITP
Italian	beginner

INTERESTS

Basketball Tennis Hiking
Bakery Graffiti

EDUCATION

Master of science engineering physics - Theoretical Physics 2025-2027
KTH ROYAL INSTITUTE OF TECHNOLOGY · Stockholm 📍
Eigenstate Thermalization Hypothesis & Random Matrix Theory · Mean field approach · Renormalization & Universality · Open quantum systems (Lindblad equation, Random unitary circuits) · Boltzmann, Fokker-Planck & Langevin equations · Symmetries in physics · Quantum field theory

General engineering school- GPA:3.9 (top 10% with honors) 2023-2027
CENTRALE LYON GENERAL ENGINEERING SCHOOL · Lyon 📍
Quantum mechanics · Problem Resolution Strategies in Python (graphs and complexity) · Graphical interface programming in C++ (see Github) · Signal processing · Chaotic Systems (MATLAB) · Nuclear engineering · Numerical Analysis and Probability theory

Preparatory class for engineering schools 2021-2023
LYCÉE DESCARTES · Tours 📍
Strong knowledge on maths, physics, chemistry, computer science · Lab works in physics and chemistry every week · Project on data transmission technologies

SCIENTIFIC PROJECTS

Simulation of many-body quantum systems (see Git) Feb.-Mar. 2026
KTH ROYAL INSTITUTE OF TECHNOLOGY · Stockholm 📍
Simulated entanglement entropy scaling in spin chains using random unitary circuits with potential · Solved Lindblad master equation for a two-level atom in a cavity to compute the population dynamic

Nonlinear Dynamics and Chaos in Complex Systems Jan. 2025-Apr. 2025
CENTRALE LYON · Lyon 📍
Computed Lyapunov exponents (Benettin algorithm) to characterize chaotic behavior · Constructed Poincaré sections and first-return maps · Investigated transitions between chaotic and periodic regimes via control parameter variation.

Research project in nanophysics Sep. 2024-Apr. 2025
INL (CNRS) · Lyon 📍
Studied Ga droplet behavior in Si-SiO₂ cavities to improve verticality of GaAs nanowires · Ran simulations on an energy minimization software, Surface Evolver

2048 Game in C++ (see Git) Jan.-Apr. 2025
CENTRALE LYON · Lyon 📍
Project carried out as part of the elective course "Graphical Interface Programming in C++"

EXPERIENCE

Internship in laboratory astrophysics May-July 2025
IRAP (CNRS) · Toulouse 📍
Measured and analyzed infrared spectra of pentagon-containing PAHs using a Fourier transform infrared spectrometer · Modeled spectra using Python/C, enabling comparison with experimental data

Teaching assistant in physics and chemistry of matter Oct. 2024-Apr. 2025
CENTRALE LYON · Lyon 📍
Tutored first- and second-year students in Physics and Chemistry of matter. Two hours once a week. Job offered following excellent grades.

Basketball instructor for adult beginners Jan 2026-Today
SSIF · Stockholm 📍
Coach paid basketball sessions for adult beginners once a week